As you can imagine, computing these ratios by hand is tedious and error prone, so I programmed my computer to find bats by first finding all peaks and valleys and then using the method described to glue them together.

Volume. I'll discuss a few volume statistics in Table 4.2, but volume trends downward in the pattern most of the time. With few samples, it's difficult to say whether up or down trending volume leads to better or worse performance, but current statistics favor a downward volume trend for the best performance.

Duration. I limited patterns to a length of 6 months or less. That's an arbitrary limitation.

Table 4.2 General Statistics

Description	**Bull Market**	**Bear Market**
Number found	537	128
Breakeven failure rate	17.7%	4.5%
Average decline after D	–14.3%	–20.2%
Volume trend	74% Downward	67% Downward
Performance Up/Down volume	–14%U, –15%D	–20%U, –20%D

Focus on Failures

Figure 4.3 shows a failure of a bearish bat pattern. Why is it a failure? As the name implies (*bearish* bat, not bullish), price should drop after point D (meaning price should trend lower after the pattern ends). However, price continues trending higher in this example. Price closes above the top of the pattern (X) at point E, posting an upward breakout.

I checked the statistics and found that 70% of the 665 bats I looked at break out upward, regardless of the market condition (bull or bear). Don't let that number give you nightmares. As awful as the 70% number is, it doesn't tell the complete story. Swing traders will soon learn (as discussed in Table 4.2) that price turns lower at point D 86% of the time. Price may still break out upward, as Figure 4.3 shows, but maybe you can make money when price drops between D and the breakout.